Recent Work

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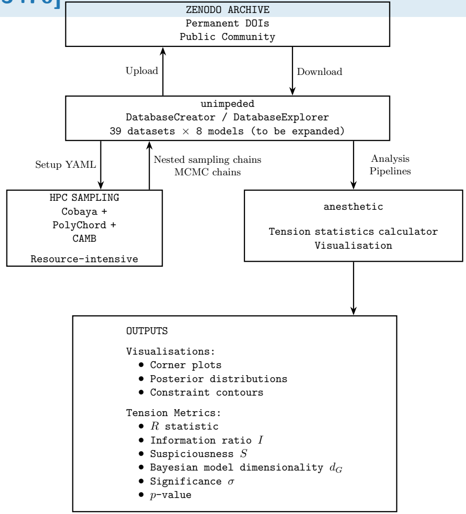
20th January 2026

unimpeded: Public Nested Sampling Database

Ong & Handley [2511.05470]

The Problem

- Model comparison needs **Bayesian evidences**
- Computing chains is expensive (DiRAC time)
- Results often not reproducible or shareable



The Solution: A Public Chain Database

- `pip install unimpeded`
- 8 cosmological models × 39 datasets
- Pre-computed NS + MCMC chains on Zenodo
- Built-in tension statistics (6 metrics)

Like Planck Legacy Archive, but with nested sampling

Normalised posteriors + evidences for everyone

`unimpeded`: Model Comparison Results

Ong & Handley [2511.04661]

Reading the Plot

- Rows: cosmological models
- Columns: datasets
- Values: $\log \mathcal{Z}$ (Bayesian evidence)
- Blue: model disfavoured
- Red: model favoured

Key Findings

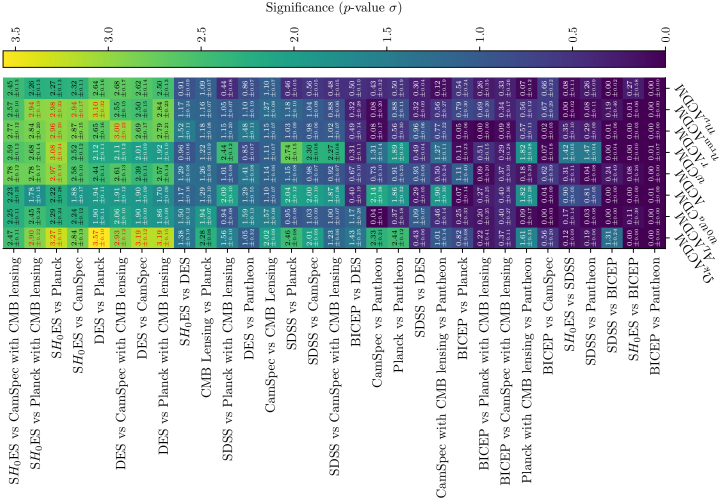
- ΛCDM consistently performs well
- Extended models penalised by Occam factor

	wCDM	w_0w_aCDM	ΛCDM	$\Omega_k\Lambda$CDM	$A_L\Lambda$CDM	$m_\nu\Lambda$CDM	$n_{\rm run}\Lambda$CDM	$r\Lambda$CDM	$\sum_j \log Z_j$
Planck	−0.97 ±0.14	−1.50 ±0.19	−3.11 ±0.21	−1.26 ±0.17	−3.07 ±0.21	−5.44 ±0.21	−7.07 ±0.22	−6.67 ±0.22	−1438.67 ±0.10
Planck with CMB lensing	−0.84 ±0.13	−0.98 ±0.15	−1.89 ±0.20	−4.46 ±0.23	−4.74 ±0.23	−4.93 ±0.23	−6.19 ±0.22	−6.35 ±0.23	−1443.91 ±0.12
CamSpec with CMB lensing	−0.90 ±0.14	−1.08 ±0.15	−1.65 ±0.17	−3.72 ±0.22	−4.62 ±0.21	−4.73 ±0.21	−6.25 ±0.22	−5.11 ±0.21	−5321.37 ±0.10
CamSpec	−0.97 ±0.13	−1.23 ±0.15	−1.97 ±0.19	−2.08 ±0.19	−3.14 ±0.20	−5.02 ±0.22	−6.25 ±0.20	−5.77 ±0.21	−5316.35 ±0.10
DES	−2.99 ±0.14	−3.38 ±0.15	−1.84 ±0.12	−1.35 ±0.11	−1.80 ±0.13	−2.22 ±0.14	−2.94 ±0.14	−1.82 ±0.12	−277.51 ±0.06
BICEP	−1.98 ±0.08	−2.23 ±0.09	−1.70 ±0.08	−1.81 ±0.08	−1.74 ±0.07	−1.96 ±0.08	−2.46 ±0.09	−5.77 ±0.11	−280.12 ±0.03
SH_0ES	−3.33 ±0.09	−4.06 ±0.09	−1.83 ±0.08	−2.10 ±0.08	−1.95 ±0.08	−1.60 ±0.07	−1.91 ±0.08	−1.78 ±0.08	−527.39 ±0.03
CMB Lensing	−2.10 ±0.07	−2.39 ±0.08	−1.63 ±0.07	−2.23 ±0.08	−2.05 ±0.07	−1.81 ±0.07	−3.73 ±0.09	−1.81 ±0.07	−11.15 ±0.03
SDSS	−1.93 ±0.07	−3.22 ±0.08	−1.76 ±0.06	−2.92 ±0.08	−1.74 ±0.06	−2.40 ±0.07	−1.84 ±0.06	−1.82 ±0.06	−16.45 ±0.03
Pantheon	−3.32 ±0.05	−3.99 ±0.06	−1.83 ±0.04	−2.08 ±0.04	−1.78 ±0.04	−1.86 ±0.04	−1.76 ±0.04	−1.82 ±0.04	−519.52 ±0.02

$\log \mathcal{P}(\mathcal{M}_i|D)$: −1, −2, −3, −4, −5, −6, −7

Ong & Handley

[2511.04661]



PolySwyft: Sequential Simulation-Based Nested Sampling

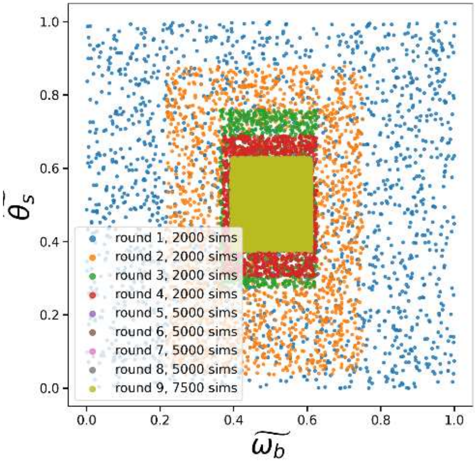
Scheutwinkel, Handley, Weniger, de Lera Acedo [2512.08316]

The Problem

- Likelihood often **intractable** (simulators only)
- Neural Ratio Estimation (NRE) learns likelihood ratios
- But: prior truncation struggles with **multimodality**

The Solution: Nest NRE inside NS

- Use nested sampling to explore full likelihood-to-evidence space
- Retrain NRE on dead points (active learning)
- KL-divergence criterion for convergence



PolySwyft: The NSNRE Algorithm

Scheutwinkel, Handley, Weniger, de Lera Acedo [2512.08316]

NSNRE

1. Initialise Sample from prior to prepare initial training dataset for the NRE r_0^*: $\{\theta_n\}_0 \sim \pi(\theta)$

2. Simulate
Generate data using a simulator: $\{\theta_n\}_i \to \mathrm{Sim}(\theta) \to \{D_n\}_i$. Create disjoint distribution $\pi \times Z$ by shuffling the pairs in joints $(\theta_n, D_n) \to (\theta_j, D_k)$ where $j \neq k$ and concatenate with joints to generate a training dataset $\{(\theta, D)\}_{i,\mathrm{train}}$.

3. NRE
Concatenate the newly simulated training dataset $\{(\theta, D)\}_{i,\mathrm{train}}$ with the previous cumulative training dataset $\{(\theta, D)\}_{0:i-1,\mathrm{train}}$. Train Neural ratio estimator $\log r_i^*$ with weights initialised from the previous run to resume training.

4. NS
Run nested sampling on $\log r_i^*(\theta, D_{\mathrm{obs}})$ to generate $\{\theta_n\}_i$ dead samples between prior and posterior targeted at D_{obs}.

5. Terminate if $\mathcal{D}_{\mathrm{KL}}$ has converged